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Editors
Journal of Classification

Dear Editors,

Please consider the manuscript “Spherical Splits for Decision Trees and Random Forests: A Short Note” for publication in the Journal of Classification as a short note.

The manuscript proposes and evaluates a simple geometric extension of tree-based classification and regression models: replacing some axis-aligned threshold questions by hyperspherical membership questions. I believe the note fits the Journal of Classification because it concerns classification methodology, tree-structured models, statistical learning, and the geometry of recursive partitions. The contribution is intentionally compact: it defines the split rule, records several useful theoretical properties, notes that CART cost-complexity pruning transfers directly, and reports exploratory evidence that spherical random forests can improve mean classification performance on tabular benchmark datasets.

The manuscript is original work, has not been published previously, and is not under consideration elsewhere. To respect the journal’s double-blind review policy, the main manuscript is anonymized. Author details and declarations are supplied on a separate title page. A blinded reproduction document and supplementary code/results are prepared for review; the public repository is <https://github.com/MathiasValla/spherical-decision-trees> and can be disclosed in the published version.

Sincerely,

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